

Using The HeartRateLab.jl for Normative Evaluation of Heart Rate Variability from Open Datasets

Alberto Barradas (abcsds) — Heart Rate Variability analysis, entirely in Julia



github.com/abcsds/HeartRateLab.jl

Abstract

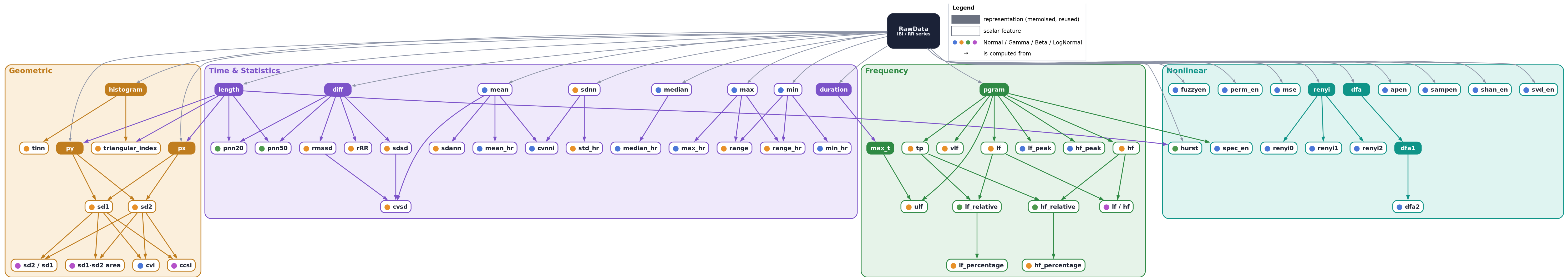
A common method in the evaluation of human biosignals is comparing experimental signals against datasets of normative values. Thanks to the open scientific resources for heart rate analysis, and the resilient network of scientific programming functionality that the Julia community enables, the normative analysis of Heart Rate Variability (HRV) can be done **completely in Julia**. By contrasting features in windows of **360 heart beats** against normative datasets we find that, during meditation, practitioners from the *Heart Rate Oscillations during Meditation* dataset (PhysioNet) show modest increases in **pNN50** ($z = +0.84$) and **SDNN** ($z = +0.81$), while **Mean IBI** ($z = +0.23$) and **RMSSD** ($z = +0.38$) stay near the healthy median — a mild, coherent vagal shift within normal range. Results demonstrate how correctly implemented open science can answer contextual questions such as: *“Is this new experimental data normal?”*

Why context matters

We contrast a time series of **Inter-Beat Intervals (IBIs)** against the feature distributions of existing open datasets, emphasising the **contextual nature** of scientific inquiry. This capability is a novel addition implemented specifically for HeartRateLab.jl.

An example contrasting meditation recordings demonstrates current capabilities and illustrates functionality that aids **hypothesis testing**, **feature engineering**, and a deeper understanding of recordings — including use as a **personal daily HRV monitor** that norms your own signals against a healthy population.

The feature computational graph — 53 features computed from each other

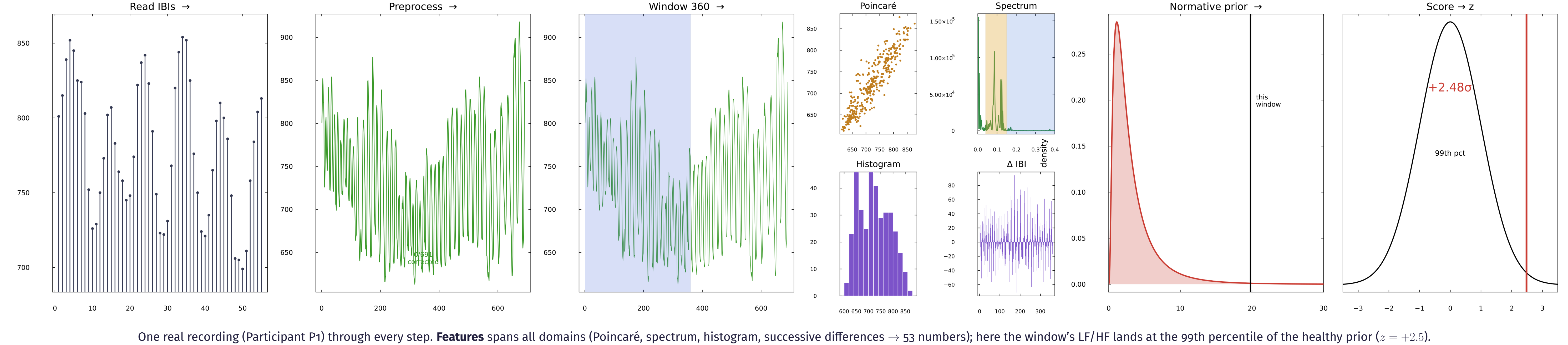


HRV features are **not independent measurements** — one IBI series feeds **11 memoised representations** which feed **53 scalar features**; every arrow is an “is computed from” dependency read from the live Features.jl registry. RMSSD, SD1 and SDSD are the *same* quantity up to a constant; LF/HF, LF% and LF-rel are all functions of LF and HF — reporting them as independent evidence multiplies one fact, the **definitional** root of the empirical redundancy the clustermap shows below.

Method — one reproducible pipeline, end to end in Julia

Read IBIs → **preprocess** (bio-outliers, ectopic & zero beats) → **53 HRV features** (time / frequency / geometric / nonlinear) → **window** (360 beats / 120 stride) → fit a **normative prior** per feature (Normal / Gamma / Beta / LogNormal) by MLE over **56,472 windows from 66 healthy subjects** → **score**: percentile = $F_{\text{prior}}(x)$, $z = \Phi^{-1}(\text{percentile})$.

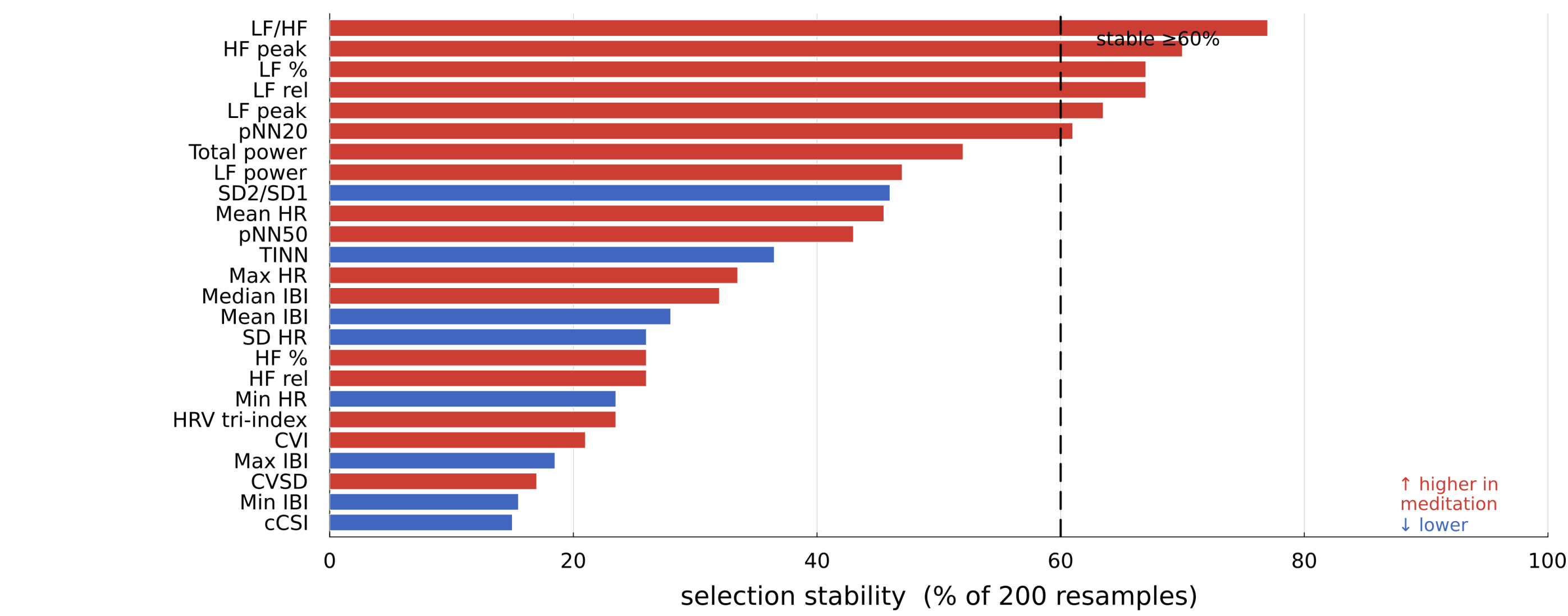
Scope. Priors are MLE approximations (KS-rejected at $n = 56k$); a z is a percentile re-expressed on a standard normal, window-level and descriptive — not a per-subject inferential test.



One real recording (Participant P1) through every step. **Features** spans all domains (Poincaré, spectrum, histogram, successive differences → 53 numbers); here the window's LF/HF lands at the 99th percentile of the healthy prior ($z = +2.5$).

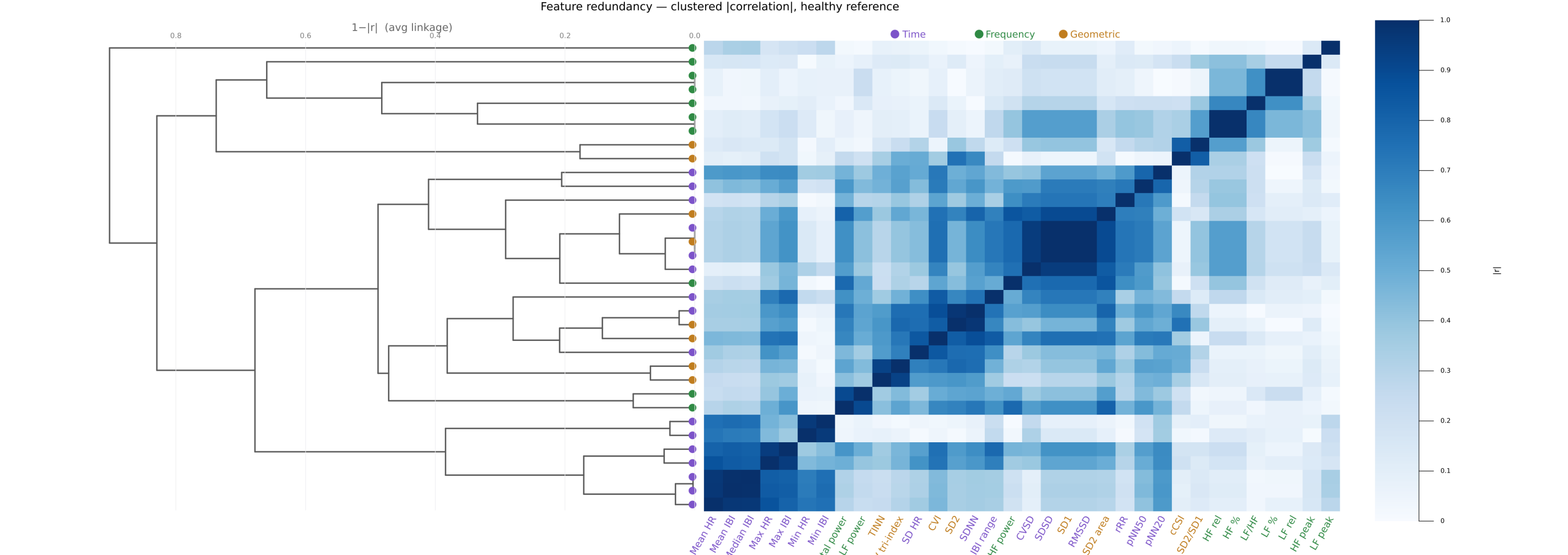
Results

Which features distinguish meditation



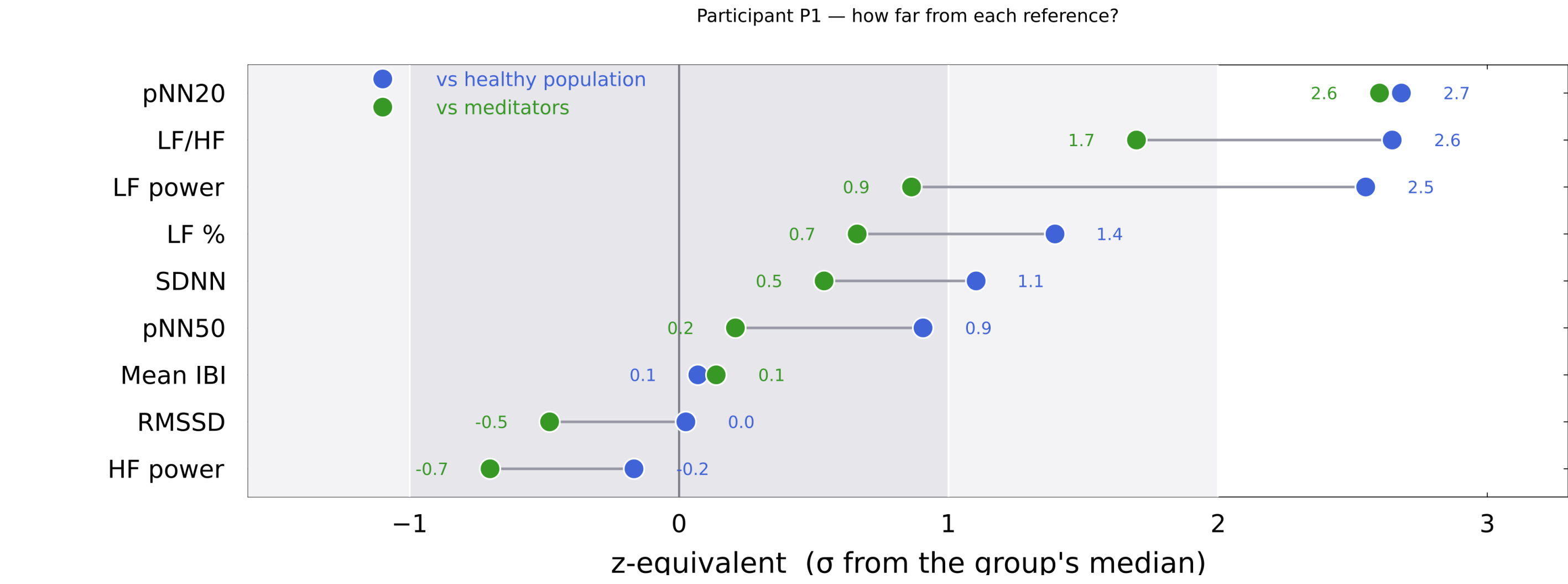
L1-logistic **stability selection** (meditation vs healthy, 200 subject-grouped resamples): the ~0.1 Hz LF-band features are selected most often. Grouped 5-fold CV **AUC** = 0.73 ± 0.27 — modest (only 11 meditators). Features are chosen by the model, not by hand.

HRV features are largely redundant



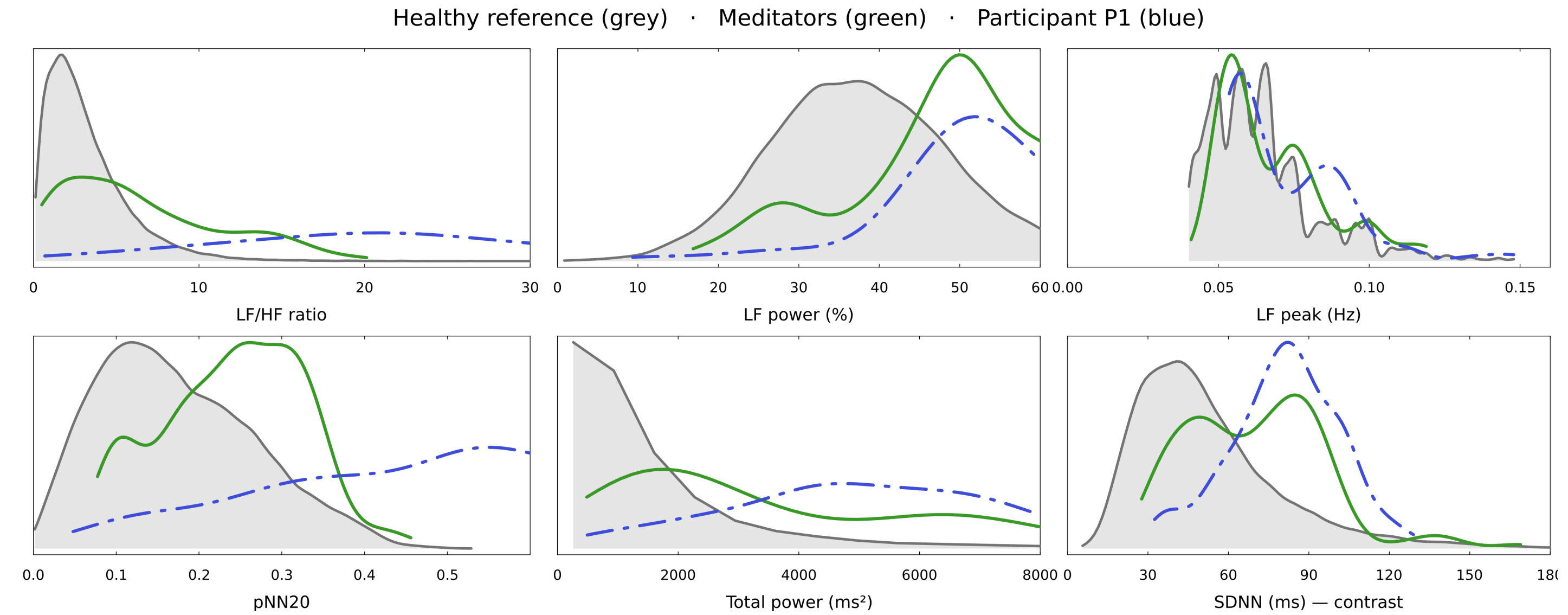
Clustered |correlation| on the healthy reference; the dendrogram groups near-duplicates. Dark blocks: SD1≡RMSSD≡SDSD ($r = 1.00$), LF%=LF-rel, SDNN≈SD2 — one variability axis wearing many names.

Participant P1 scored against two references



P1's deviation (σ -equivalent) from the **healthy population** and from the **meditators**. On the LF-band features P1 is extreme vs the general population (LF/HF $+2.6\sigma$) but close to a typical meditator ($+1.7\sigma$) — the line is the gap between the two references.

Distributions: population, meditators, participant



Grey = healthy reference (66 subjects); **green** = meditators (11 practitioners); **blue** = P1. Both meditators and P1 shift the LF-band features right of healthy; short-term time-domain spread (SDNN) separates them much less.

“Is this new experimental data normal?” HeartRateLab.jl answers it **in context** — open-data priors and a principled, reproducible pipeline. Meditation ($n = 263$ windows): LF-band $\uparrow$, discrimination modest. Participant P1 ($n = 1$): LF-band strongly $\uparrow$ vs the population, meditator-like. **Limitations:** priors are KS-rejected approximations; windows are correlated (descriptive, not inferential); meditation $n = 11$; P1 is a single self-tracked participant. Full detail + the 44-feature table in the notes handout.